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$$V_s = V_m \sin(\omega t)$$

$$V_o = |V_m \sin \omega t|$$

$$KVL: L \frac{di}{dt} + Ri = |V_m \sin \omega t|$$

$$\Rightarrow L \frac{di}{dt} + Ri = V_m \sin \omega t \quad \Sigma 0, \pi \} \rightarrow i = k e^{-\frac{R}{L}t} + \varphi \sin(\omega t - \phi)$$

$$\rightarrow \phi = \tan^{-1}\left(\frac{\omega L}{R}\right), \quad \varphi = \frac{V_m}{\sqrt{R^2 + (\omega L)^2}}$$

$$k - \varphi \sin(\theta) = k e^{-\frac{R}{\omega L} \pi} + \varphi \sin(\pi - \theta) \Rightarrow k = \frac{\varphi \sin \theta}{1 - e^{-\frac{R\pi}{\omega L}}}$$

! تو به رابطه بدست آمده از رابطه (I) ساده نخواهد بود پس باید اگر چه استقیم

آن را از روی $\sin$ و $\cos$ از آن مشتق بگیریم و با یکدیگر:

$$V_o = |V_m \sin(\omega t)| \Rightarrow |\sin(\omega t)| = a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega t) + \sum_{n=1}^{\infty} b_n \sin(n\omega t)$$

$$\rightarrow a_0 = \frac{1}{\pi} \int_0^{\pi} \sin(\omega t) d\omega t = \frac{1}{\pi} [-\cos(\omega t)]_0^{\pi} = \frac{2}{\pi} \quad \text{(II)} \quad \text{(I)}$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} \sin(\omega t) \cos(n\omega t) d\omega t = \frac{2}{\pi} \frac{1 - \cos(n\pi)}{1 - n^2} \quad \text{(III)}$$

$$\text{(I) \& (II) \& (III)} : |\sin(\omega t)| = \frac{2V_m}{\pi} - \sum_{n=1}^{\infty} \frac{4V_m}{\pi(n^2 - 1)} \cos(n\omega t)$$

و

$$b_1 = \frac{4}{\pi} \frac{V_m}{\omega} , b_2 = \frac{4}{\pi} \frac{V_m}{\omega}$$

هنا، صوبہ کے لیے جیسا کہ ہم نے دیکھا ہے، اس کے لیے ہم نے ایک خاص انداز میں لکھا ہے کہ اس کے لیے ہم نے ایک خاص انداز میں لکھا ہے

$$I_{ae} = \sqrt{I_1^2 + I_2^2 + \dots} \rightarrow 1\% \text{ کا فرق}$$

$$V_o \approx \frac{2V_m}{\pi} - \frac{4V_m}{\pi^2} \cos(\omega t) / I_{o,dc} = \frac{2V_m}{\pi R} / I_{o,ac} = \frac{4V_m}{\pi^2 \sqrt{R^2 + (\omega L)^2}}$$

$$\Rightarrow I_{o,max} - I_{o,min} = I_{o,ae} = \frac{4V_m}{\pi^2 \sqrt{R^2 + (\omega L)^2}} \rightarrow \frac{\Delta I_o}{I_{o,dc}} = \frac{4}{100}$$

$$\Rightarrow R = 10 \Omega \Rightarrow L = \frac{4 \times 10^{-3}}{\omega} \text{ (ER)}$$

حل (3) (الف)

$$\left. \begin{aligned} V_a - L_1 \frac{di_1}{dt} &= V_o \\ V_b - L_2 \frac{di_2}{dt} &= V_o \end{aligned} \right\} (+) \Rightarrow V_a + V_b - L \left(\frac{di_1}{dt} + \frac{di_2}{dt} \right) = 2V_o$$

(L = L_1 = L_2)

$$i_1 + i_2 = i_{dc} \Rightarrow \frac{di_1}{dt} + \frac{di_2}{dt} = 0 \Rightarrow \frac{V_a + V_b}{2} = V_o$$

$$\int_0^{T/2} V_o dt = \int_0^{T/2} L \omega di = \omega L I_{dc}$$

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$$V_{lost(av)} = \frac{\omega L I_{dc}}{\pi} - \frac{\omega L I_{dc}}{\pi} \rightarrow V_{oc(av)} = \frac{V}{\pi} V_{LLP} - \frac{\omega L I_{dc}}{\pi}$$

$$= 2 \times 10 \times 1.14 - 0.111 = 2.189 \rightarrow V_o(av) = \frac{V_{LLP}}{\pi} \frac{1 + \cos \mu}{\pi} \quad (*)$$

$$V_a - L \frac{di_a}{dt} = V_b - L \frac{di_b}{dt} \rightarrow V_a - V_b = \omega L_s \left(\frac{di_a}{dt} - \frac{di_b}{dt} \right)$$

N_{ab}

$$\Rightarrow V_{ba} = \omega L_s \frac{di_b}{dt} \rightarrow \frac{di_b}{dt} = \frac{V_{ba}}{\omega L_s} = \frac{V_{LLP} \sin \mu \omega t}{\omega L_s}$$

$$I_{dc} = \int_0^\mu \frac{V_{LLP} \sin \omega t}{\omega L_s} d\omega t = -\frac{V_{LLP}}{\omega L_s} \cos(\omega t) \Big|_0^\mu \Rightarrow I_{dc}$$

$$= \frac{V_{LLP}}{\omega L_s} (1 - \cos \mu) \quad (*) \Rightarrow \cos \mu = 1 - \frac{\omega L_s I_{dc}}{V_{LLP}}$$

$$\Rightarrow \cos \mu = 1 - \frac{1 \times 5 \pi \times 200 \times 10^{-3} \times 500}{\sqrt{2} \times 500} = 0.747 \Rightarrow \mu = \cos^{-1}(0.747) = 41.9^\circ$$

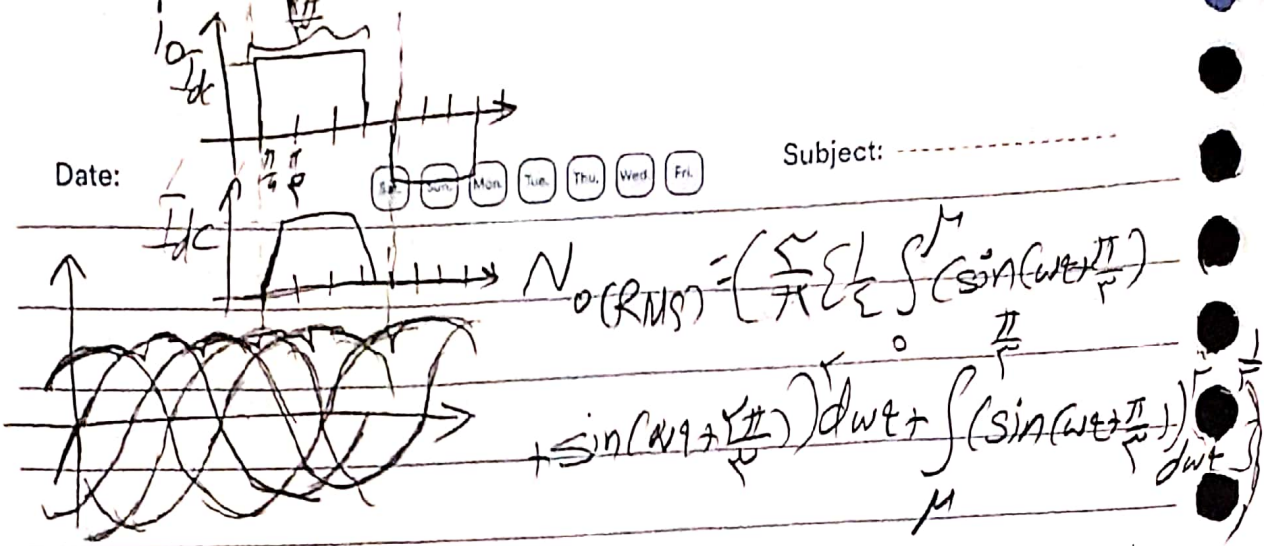
$$\Rightarrow \mu > 90^\circ \Rightarrow L = \frac{V}{\omega} \mu H \rightarrow \cos \mu = 1 - \frac{1 \times 5 \pi \times 200 \times 10^{-3} \times 500}{\sqrt{2} \times 500} = 0.111$$

$$\Rightarrow \mu = 50.1^\circ \text{ to } 119.9^\circ$$

Notch depth = $V_{LLP} \sin \mu \times 500 \sqrt{2} \sin \mu$
= 0.111

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$$V_o(RMS) = V_{dc} \left(\frac{1}{\sqrt{2}} + \frac{\sqrt{2}}{4\pi} \frac{\tau}{\tau_c} + \frac{\sqrt{2}}{4\pi} \cos(\tau/\tau_c) + \frac{\sqrt{2}}{4\pi} \sin(\tau/\tau_c) \right)$$

= 0.707 V_{dc}

$$\Rightarrow RF = \frac{V_{ac}}{V_{dc}} = \sqrt{\left(\frac{V_{rms}}{V_{dc}} \right)^2 - 1} \approx 0.45 \approx 45\%$$

$$THD = \sqrt{\frac{(I_{rms})^2}{(I_1)^2} - 1}$$

Ideal case $\Rightarrow I_{rms} = \left(\frac{1}{\pi} \int_0^\pi i(\omega t) d\omega t \right)$

$$= \left(\frac{1}{\pi} \times \frac{\sqrt{2}}{2} \times I_{dc} \right) = \sqrt{\frac{2}{\pi}} I_{dc} = 0.799 I_{dc}$$

$$A_1 = \frac{\sqrt{2}}{\pi} \int_0^\pi i(\omega t) \sin(\omega t) d\omega t = \frac{\sqrt{2}}{\pi} I_{dc} \left(-\cos \omega t \right) \Big|_0^\pi = \frac{\sqrt{2}}{\pi} I_{dc}$$

$$I_1 = \frac{A_1}{\sqrt{2}} = \frac{\sqrt{2}}{\pi} I_{dc} \Rightarrow THD$$

Ideal = 0.45 or 45%

$$I_{rms} = \frac{1}{\pi} \left(\int_0^\pi \left(\frac{I_{dc}}{\sqrt{2}} \omega t \right)^2 d\omega t + \int_\pi^{2\pi} I_{dc}^2 d\omega t + \int_0^\pi \left(\frac{I_{dc}}{\sqrt{2}} \omega t \right)^2 d\omega t \right)$$

$$= \frac{1}{\pi} \left(\frac{I_{dc}^2 \pi}{4} + I_{dc}^2 \left(\frac{\pi}{2} - \pi \right) \right) \Rightarrow I_{rms} = I_{dc} \sqrt{\frac{2}{\pi} - \frac{1}{4}} \approx 0.799 I_{dc}$$

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$$a_1 = \frac{4 \times 1 \times 1 \times 1}{\pi} \int_0^{\frac{\pi}{2}} i(\omega t) \cos \omega t d\omega t = \frac{4}{\pi} \left(\int_0^{\alpha} I_{dc} \cos \omega t d\omega t + \int_{\alpha}^{\beta} \frac{B - \omega t}{\pi} d\omega t \right)$$

$$\alpha = \left(\frac{\pi}{2} - \mu \right), \beta = \left(\frac{\pi}{2} - \mu \right) + \mu$$

$$I_1 = \frac{a_1}{\sqrt{2}} = \frac{\sqrt{2}}{\pi} I_{dc} \frac{\sin\left(\frac{\pi}{2}\right)}{\frac{\pi}{2}} = 0.444 I_{dc}$$

$$THD = \frac{0.544}{1} = 54.4\% \rightarrow \text{ملاحظة: } THD \text{ اندازة قریباً ۱۰۰٪ است}$$

Commutation

سوال ۹: الف

$$\text{Reverse relative Peak Voltage} \Rightarrow V_{RRM} = V_{LL} = E_{00} \sqrt{2}$$

$$\text{Full-cycle average Voltage Current} \Rightarrow I_F(\text{avg}) = \frac{\sqrt{2}}{\pi} \frac{V_{LL}}{R}$$

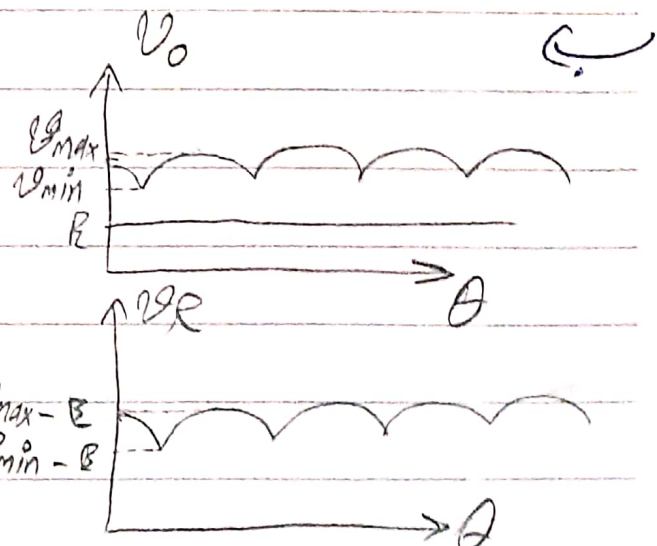
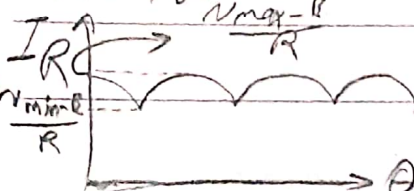
$$\frac{1}{\pi} \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} V_m \sin \omega t d\omega t = \frac{\sqrt{2}}{\pi} V_{LL}$$

$$V_{min} \approx E_{A, 19}$$

$$V_{max} \approx 290, V$$

$$E = 200$$

$$R = 0.2 \Omega$$



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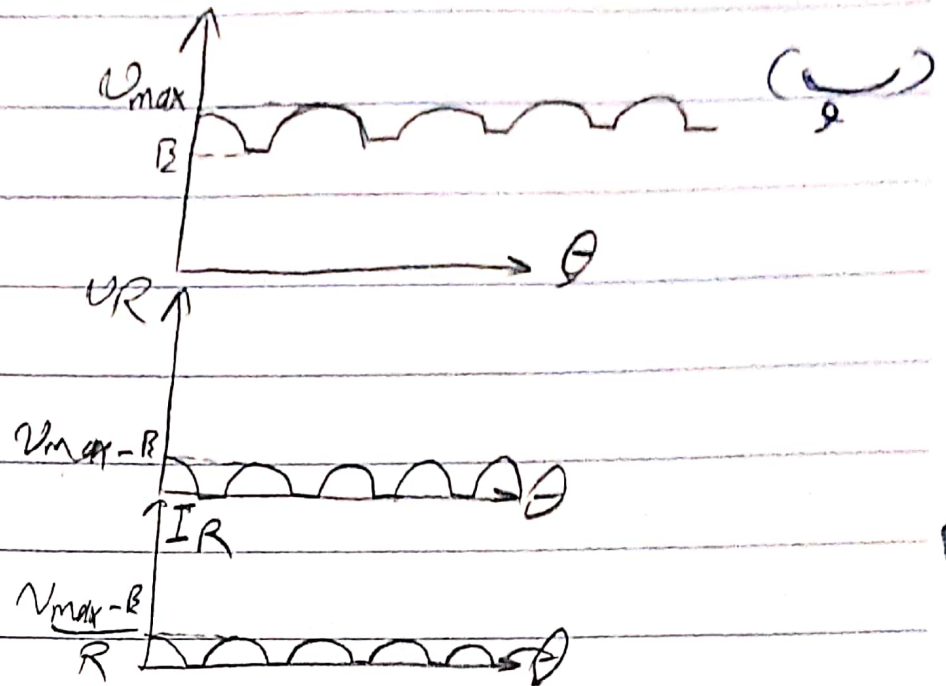
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$$\nu_{\min} \approx 219,9 \text{ }^{\circ}\text{O}$$

$$U_{\max} \approx 592, V$$

$$E = 25\% \text{ } ^{19}$$

$$R = 0.1 \Omega$$



الف: دو گونہ تاسیخ ہے۔ ۱۔ تحریر میں فرق دارد (محرر و مکتوب کے درمیان)

کتاب کیوں کر مع آفاق کا فائدہ دے گی۔ یہ سب اور کمال کا از رو اور اصل کا مقابلہ آئندہ کریں۔ یہ جملہ و کمال کا
نامتو اور مستند

علاء الدین علی ابن یحییٰ ادریس کاتبی حیدر یحییٰ زاده کابل دارو و طب و سحر

بیتہ باب مدت و حال و طبع تیرہ سال کو درمختص باچھا علم حاصل کیا اور مختلف اساتذہ

رسیدہ خزانہ